

Department of Statistics and Computer Science
University of Peradeniya
CSC2021 - Programming using Data Structures
Lab Sheet 03
QUEUES

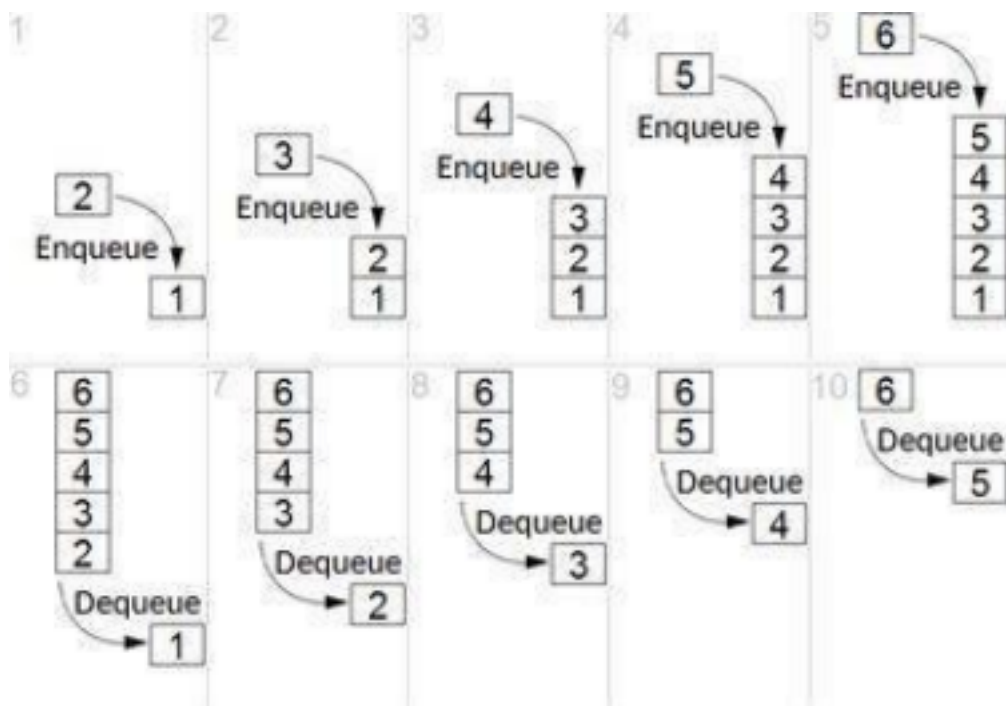
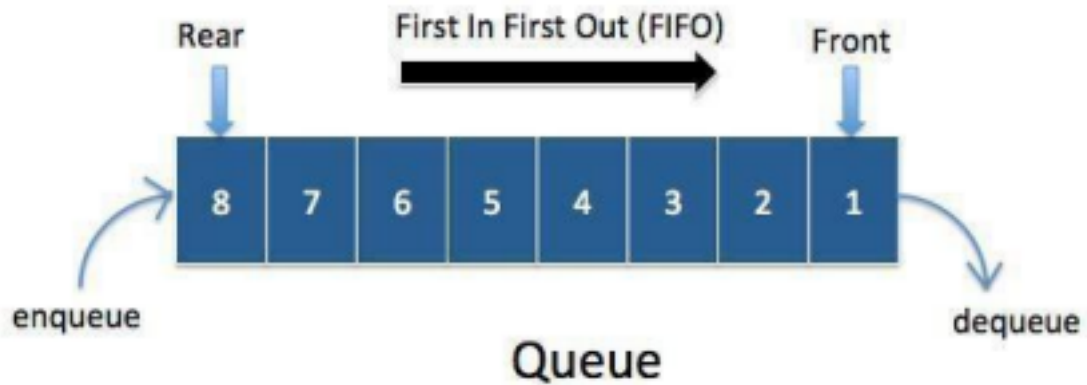
Purpose

- Learning the implementation of Queue data structure using linked Lists.
- Understanding the instance where practical applications of a queue are used.

Theory

- Based on the First In First Out (**FIFO**) algorithm.
- One of the main data structures that has been derived by Linked Lists (usually singly linked list implementation is used).
- A queue is derived by restricting the inserting and deleting operations to each end of the List. • Hence a queue can be implemented using an array as well (fixed length queue) • Inserting is restricted to be from the **end** of the queue and the insertion operation is called **enqueue/ put**.
- Retrieval is restricted to be from the **start** of the queue and the retrieval operation is called **dequeue/ get**.
- Queues also have two links/ references in their structure which are called **front** and **back**. • Basic Operations of a queue include.

- Put - Insert new elements to the end of the queue
 - Peek - Retrieve data component of the first element at the start of the queue without removing the element
 - Get - Retrieve elements from the start of the queue. (Return the data component from the node and remove the node from the list).
 - Display - Print the elements of the nodes currently in the queue starting from the beginning.
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- Priority Queue is a Special type of a queue where enqueueing (insertion) is done according to a specific pre-declared priority (The elements of the priority queue can be ordered according to the natural ordering, or by a declared *Comparator method*- similar to an insertion to a sorted linked list).
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- Visual Implementation of a basic queue,



Question 01

1. Implement the Queue Data Structure by implementing the functions mentioned above
(Assume the data section of the node contains only one integer variable).

a. In the Driver class do the following,

i. Create a new Queue and insert the following values,

• 2, 5, 38, 44, 73, 105, 225, 515

ii. Retrieve a value from the queue

iii. Insert 30, 10

iv. Retrieve another value from the queue and insert 838 and 586.

v. Print the contents of the queue.

Question 02

1. There is a vehicle service center where the access road to the center can accommodate any number of vehicles at a given time. The center is built similar to a processing line of a factory in such a way that the vehicle that comes first will be serviced first. Each of the vehicles are identified by Their Vehicle Identification number (VIN). Apart from the VIN, Vehicle Type (eg: Car, Van, SUV, Jeep, etc.), and Service type (eg: Full Service, Body wash only, Interior Cleaning only, Oil Change only, etc.) should be stored as service details. Write a program to handle the vehicle accommodations in the garage, allowing the following commands,
 - a. **inLine** - checks whether a vehicle is in the access road for servicing. If the vehicle is in the line this method should print out how many vehicles should be serviced before the searched vehicle is to be serviced (Assume that the first vehicle in the list is the one currently being serviced).
 - b. **enterForService** - Adds a vehicle to the list of vehicles to be serviced.
 - c. **exitService** - Tries to move a given vehicle out of the service center if it is not at the front of the line (The vehicle being serviced), print out an error mentioning The vehicle is not yet serviced.
 - d. **showQueue** - Prints all the details of the vehicles in the service center.